SARAS Individualised Project Plan — Sushank Jung Rawal

Faculty: Faculty of Technology, School of Computer Science
Module: CET257 Enterprise Project
Module Leader: Dr Becky Allen
Assessment: Assignment 1: Group Assignment (30% of Module Total)
Task Deliverable: Task A5 — Initial Project Plan (10 Marks / 10%)
Sub-Component: 03 — Individualised Project Plan: Sushank Jung Rawal
Assigned Student: Sushank Jung Rawal (Deputy Project Manager & Frontend UI/UX Lead)
Partner: Rijan Kapur Poudel (Team Lead / Project Manager & Technical Architecture Lead)
Client: Mr. Ninson Lama, Autoport Pvt. Ltd. (`ninsonlama@gmail.com`)
Primary Working Workbook: `Project Plan.xlsx` (Worksheet: `Sushank Individual Plan`)

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1. Executive Role Profile & Diagnostic Competency Baseline

1.1 Role Designation & Functional Leadership

- **Primary Role:** Deputy Project Manager & Frontend UI/UX Lead
- **Secondary Specialisms:** 360° Spatial Interaction Engineer, Accessibility (A11y) Specialist, Brand & Creative Director
- **Functional Remit:** Responsible for user-centred design research, PACT user analysis, Nielsen usability heuristic evaluations, responsive UI component architecture (Next.js 15, React 19, Tailwind CSS), 360° panoramic spatial inspection scenes (Photo Sphere Viewer 5), WCAG 2.2 accessibility compliance, corporate brand identity, promotional website design, and presentation slide design.

1.2 HEA Skills Audit Diagnostic Baseline

In accordance with the diagnostic self-assessment completed in Week 1 using the official Higher Education Academy (HEA) Student Employability Profile framework (`Sushank Jung Rawal - Skills Audit.docx`), the following capability baseline was established:

- **Demonstrated Core Strengths:**
  - *UI/UX Wireframing & Design Systems:* Advanced capability in translating complex procedural workflows into accessible, multi-device wireframes and establishing tokenized CSS design systems.
  - *Modern Semantic Web Technologies:* Strong proficiency in HTML5, CSS3, modern layout models (Flexbox, CSS Grid), Tailwind CSS utility tokens, and React component modularity.
  - *Corporate Brand & Visual Communication:* Demonstrated expertise in brand mark geometry, visual hierarchy, typography selection, palette balance, and multimedia slide deck presentation design.
  - *Human-Computer Interaction (HCI) Frameworks:* Academic understanding of usability engineering principles, user mental models, and structured cognitive walkthroughs.
- **Targeted Professional Development Areas:**
  - *WebGL Spatial Rendering & 360° Panorama Integration:* Identified a diagnostic goal to develop hands-on technical proficiency with browser-based WebGL panorama renderers (Photo Sphere Viewer 5) and equirectangular spatial coordinate systems.
  - *Advanced Web Accessibility (WCAG 2.2):* Requirement to master rigorous accessibility engineering, specifically implementing non-dragging keyboard alternatives for complex interactive sequencing activities (Criterion 2.5.7).

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2. Allocated Project Responsibilities & Governance Remit

To eliminate operational ambiguity and support the two-person team structure, Sushank's responsibilities were allocated across six distinct functional domains:

1. **User Requirements Engineering & HCI Analysis:** Lead user research using the PACT framework (People, Activities, Contexts, Technologies), establish user personas, define user stories, and conduct formal Nielsen 10 Usability Heuristic reviews.
2. **Frontend UI Architecture & Component Engineering:** Develop the interactive step-by-step guided lesson wizard (`lesson-experience.tsx`), responsive application containers, and formative learning activity renderers (classification, matching, sequencing).
3. **360° Spatial Inspection Integration:** Mount equirectangular warehouse panorama scenes using Photo Sphere Viewer 5, configure interactive hazard hotspots, and optimise WebGL texture loading performance on constrained viewports.
4. **Accessibility Hardening (WCAG 2.2):** Ensure full keyboard navigation operability, visible focus indicators, screen reader accessibility attributes (ARIA), and accessible button reordering alternatives to drag-and-drop interactions.
5. **Corporate Brand Identity & Visual Assets:** Design the SARAS crane brand mark, corporate identity manual, semantic design token architecture (standardising on SARAS Primary Blue `#0056D2`), and promotional marketing wireframes.
6. **Presentation Visual Design & Demonstration Delivery:** Design the 13-slide widescreen client presentation deck, formulate the timed rehearsal script, co-lead the live prototype demonstration, and defend design decisions during client Q&A.

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3. Master Work Package Allocation Ledger (Sushank Jung Rawal)

The table below catalogs all 14 discrete work packages allocated to Sushank Jung Rawal across Weeks 1–7. Dates and effort allocations reflect the planning baseline recorded in `Project Plan.xlsx` :

Task ID	Work Package Title & Scope	Functional Domain	Planned Start	Planned End	Planned Effort	Dependencies	Key Deliverable Output	Week 7 Status
S-01	Design system wireframes & Tailwind CSS token architecture	UI/UX Foundation	2026-07-20	2026-07-28	14 hrs	—	Design system tokens; component specs	**Completed**
S-02	Skills audit reflection on frontend engineering & visual communication	A2 Roles/Rules	2026-07-20	2026-07-28	8 hrs	INIT-01	`Sushank Jung Rawal - Skills Audit.docx`	**Completed**
S-03	SARAS brand visual identity, crane logo formulation & palette tokens	Corporate Brand	2026-07-28	2026-08-18	18 hrs	S-01	Logo PNG/SVG; brand guidelines manual	**Completed**
S-04	Step-by-step lesson wizard component (`lesson-experience.tsx`)	Frontend Eng	2026-07-27	2026-08-04	16 hrs	DEV-01	Interactive lesson wizard component	**Completed**
S-05	Interactive activity renderers (classification, matching, sequencing)	Instructional UI	2026-08-03	2026-08-12	16 hrs	S-04	Activity components; feedback banners	**Completed**
S-06	PACT User Analysis & Nielsen 10 Heuristic Usability Review	A1 User Analysis	2026-07-27	2026-08-14	16 hrs	S-01	`02 User Requirements Analysis.pdf`	**Completed**
S-07	Photo Sphere Viewer 5 integration & equirectangular panorama mounting	360° Interaction	2026-08-05	2026-08-14	18 hrs	S-04	`panorama-viewer.tsx` mounting engine	**Completed**
S-08	WCAG 2.2 accessible button reordering & reduced motion controls	Accessibility	2026-08-08	2026-08-11	10 hrs	S-05	Non-drag Move Up/Down buttons (57b7924)	**Completed**
S-09	Semantic colour token overhaul: restricting brand red to danger	UI Polish	2026-08-14	2026-08-16	8 hrs	S-03	Token refactoring; Primary Blue (#0056D2)	**Completed**
S-10	Interactive 360° hotspot inspection cards & spatial coordinates	Interactive Content	2026-08-15	2026-08-21	14 hrs	S-07	4 Fire Safety interactive hotspots	**Completed**
S-11	Zero-budget digital marketing strategy & promotional wireframes	A6 Marketing	2026-08-15	2026-08-27	14 hrs	S-03	`Marketing Website and Corporate Identity.pdf`	**Completed**
S-12	13-slide PowerPoint client progress presentation deck formulation	A7 Presentation	2026-08-18	2026-08-25	16 hrs	S-03, S-11	`saras-prototype-presentation.pptx`	**Completed**

Task ID	Work Package Title & Scope	Functional Domain	Planned Start	Planned End	Planned Effort	Dependencies	Key Deliverable Output	Week 7 Status
S-13	Prepare the two-person rehearsal script for ~15-minute presentation & demo	A7 Presentation	2026-08-24	2026-08-26	8 hrs	S-12	Timed rehearsal script (14m 55s target)	**Completed**
S-14	Deliver Week 7 client progress presentation: PACT, UI/UX, and 360° demo	Milestone 1	2026-08-31	2026-08-31	6 hrs	S-13	Formal presentation delivery; Q&A defense	**Delivered**

4. Chronological Week-by-Week Planned vs. Actual Activity Ledger

Week 1 (20 July – 26 July 2026): Governance Setup & Visual Foundations

- **Planned Commitments:**
  - Configure team Microsoft Teams workspace with structured operational channels.
  - Establish GitHub code repository and configure branch protection guidelines.
  - Complete personal diagnostic skills audit using the official HEA template.
  - Draft initial wireframe concepts and Tailwind CSS design token baseline.
- **Actual Execution & Outputs:**
  - Configured Microsoft Teams with dedicated channels for Client Liaison, Technical Dev, Governance, and Design.
  - Initialized Git repository; configured commit conventions.
  - Completed `Sushank Jung Rawal - Skills Audit.docx` highlighting UI/UX proficiency and WebGL development goals.
  - Established foundational design tokens for typography, container spacing, and responsive breakpoints.
- **Variance & Corrective Action:** None. Tasks completed on schedule.

Week 2 (27 July – 02 August 2026): User Requirements & Lesson Wizard UI

- **Planned Commitments:**
  - Execute PACT user requirements analysis decomposing logistics worker operating contexts.
  - Build the step-by-step interactive lesson wizard (`lesson-experience.tsx`).
  - Ensure responsive container layouts supporting desktop, tablet, and mobile viewports.
- **Actual Execution & Outputs:**
  - Completed PACT framework analysis; identified cognitive load limits for floor operatives.
  - Developed `lesson-experience.tsx` wizard component featuring clear step progression, back/continue navigation, and explanatory feedback panels.
  - Implemented responsive CSS Grid and Flexbox layouts verified on mobile (375px) and desktop (1280px).
- **Variance & Corrective Action:** Incorporated mobile-first responsive breakpoints early to ensure tablet compatibility for floor supervisors.

Week 3 (03 August – 09 August 2026): Formative Activities & Client Consultation

- **Planned Commitments:**
  - Develop multi-type activity renderers for formative safety practice.
  - Attend formal client consultation with Mr. Ninson Lama on 06 August.
  - Begin technical investigation into Photo Sphere Viewer 5 browser mounting.

- **Actual Execution & Outputs:**
  - Implemented interactive components: classification cards, matching pairs, and sequencing lists.
  - Participated in client consultation; captured client directive that incorrect responses must provide clear, actionable explanations rather than bare pass/fail indications.
  - Upgraded feedback alert components to display rich didactic explanations.
- **Variance & Corrective Action:** Re-engineered activity feedback modals to provide educational explanations following client guidance.

Week 4 (10 August – 16 August 2026): 360° Scene Mounting & Accessibility Engineering

- **Planned Commitments:**
  - Integrate Photo Sphere Viewer 5 with equirectangular warehouse panorama.
  - Conduct comprehensive accessibility review against WCAG 2.2 standards.
  - Resolve keyboard navigation traps and tab-order anomalies.
- **Actual Execution & Outputs:**
  - Mounted Photo Sphere Viewer 5 inside React client shell (`panorama-viewer.tsx`).
  - Conducted accessibility audit; discovered that drag-and-drop sequencing violated WCAG 2.2 Criterion 2.5.7 (Dragging Movements) for keyboard-only and motor-impaired users.
  - Engineered accessible 'Move Up' and 'Move Down' button alternatives alongside drag handles (Commit `57b7924` , `fa280e7`).
- **Variance & Corrective Action:** Unplanned development of non-drag alternative controls required 4 engineering hours; delivered on time without delaying milestone delivery.

Week 5 (17 August – 23 August 2026): Spatial Hotspots, Brand Tokens & Marketing Strategy

- **Planned Commitments:**
  - Deploy 4 interactive hazard hotspots in the 360° Fire Safety scene.
  - Overhaul brand color tokens to eliminate usability ambiguity.
  - Draft zero-budget B2B digital marketing strategy and website wireframes.
- **Actual Execution & Outputs:**
  - Configured 4 spatial hotspots (blocked exit, missing extinguisher, uninspected electrical panel, accumulated combustibles) with rich inspection modal cards.
  - Refactored brand design tokens to enforce SARAS Primary Blue (`#0056D2`) for primary actions, strictly reserving red for physical danger/error.
  - Authored zero-budget marketing strategy document and designed promotional website wireframes.
- **Variance & Corrective Action:** Shifted brand palette away from heavy red usage to prevent user cognitive confusion with safety hazard warnings.

Week 6 (24 August – 30 August 2026): Presentation Design & Timing Optimization

- **Planned Commitments:**
  - Design 13-slide widescreen client presentation deck in PowerPoint.
  - Draft comprehensive presenter notes and formulation of two-person rehearsal script.
  - Coordinate timed dry-run rehearsals with Rijan to achieve 15-minute target.
- **Actual Execution & Outputs:**
  - Designed 13 high-impact slides adhering strictly to the SARAS visual identity and typography standards.
  - Formulated a joint rehearsal script budgeted at exactly 14 minutes 55 seconds (9m 35s slides + 5m 20s live demo), reserving over 5 minutes for client questions.
  - Conducted three timed rehearsals; refined slide handovers and demonstration screen-shares.

- **Variance & Corrective Action:** Cut 5 auxiliary concept slides during rehearsals to ensure strict adherence to the 20-minute university presentation ceiling.

Week 7 (31 August – 06 September 2026): Milestone 1 Delivery & Documentation Lock

- **Planned Commitments:**
  - Co-deliver Milestone 1 client presentation and live 360° prototype walkthrough on 31 August.
  - Lead problem framing, user analysis, design system, and spatial navigation segments.
  - Finalise all visual assets and documentation across folders A6 and A7.
- **Actual Execution & Outputs:**
  - Co-delivered presentation to Mr. Ninson Lama; smoothly navigated the live 360° Fire Safety warehouse scene; demonstrated hazard hotspot inspection.
  - Participated in client Q&A regarding spatial learning ergonomics and mobile access.
  - Consolidated marketing documents, Figma wireframe exports, presentation decks, and meeting minutes into locked submission folders.
- **Variance & Corrective Action:** Successfully completed live presentation within the allocated time window, leaving 5 minutes for comprehensive client discussion.

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5. Individual Risk Mitigation & Usability/Accessibility Management

Sushank exercised primary ownership over the project's usability, accessibility, and visual performance risks:

- **Mitigation of RSK-03 (Accessibility Exclusion — Severity 12):**
  - *Threat:* Drag-and-drop sequencing excluded operatives relying on keyboard-only navigation or assistive pointing devices, failing WCAG 2.2 accessibility compliance.
  - *Engineered Action:* Implemented accessible 'Move Up' and 'Move Down' button controls alongside drag handles (Commit `57b7924`, `fa280e7`). Added visible focus rings, ARIA live region announcements, and keyboard shortcuts, fully satisfying WCAG 2.2 Criterion 2.5.7.
- **Mitigation of RSK-05 (Brand Colour Ambiguity — Severity 9):**
  - *Threat:* Extensive use of brand red across standard UI action buttons caused cognitive interference, leading learners to confuse navigation controls with emergency stop or hazard warning alerts.
  - *Engineered Action:* Standardised primary UI actions on SARAS Primary Blue (`#0056D2`), reserving red exclusively for physical hazards, incorrect assessment feedback, and critical system errors.
- **Mitigation of RSK-06 (360° Texture Latency — Severity 9):**
  - *Threat:* Large equirectangular WebGL textures caused mobile browser memory pressure, resulting in dropped animation frames and slow load times.
  - *Engineered Action:* Compressed 360° panorama textures to optimised JPEG formats; integrated progressive CSS loading spinners and static fallback inspection cards for constrained viewports.

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6. Professional Skills Development & HEA Competency Reflection

The execution of Weeks 1–7 provided structured experiential learning directly addressing the developmental priorities identified in the initial skills audit:

- **Mastery of WebGL Spatial Interfaces:** Gained practical engineering expertise in mounting, manipulating, and optimizing 360-degree equirectangular environments within React/Next.js single-page applications using Photo Sphere Viewer 5.
- **Advanced Accessibility Compliance (WCAG 2.2):** Deepened technical knowledge of modern accessibility standards, transitioning from basic semantic HTML to advanced ARIA state management and motor accessibility

patterns.

- **Enterprise Presentation Pacing & Design:** Mastered the discipline of timing-constrained corporate communication, refining an extensive 40-minute conceptual vision into a crisp, high-impact 15-minute presentation that earned positive client engagement.

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7. Forward Trajectory: Weeks 8–12 Individual Deliverables

During Phase 2 (Enterprise Hardening & Final Product Handover), Sushank's allocated responsibilities comprise:

1. **Secondary Module Scene Integration (Weeks 8–9):** Capture and configure equirectangular panorama scenes for the **Hazard Perception** and **Manual Handling** modules.
2. **Supervisor Dashboard UI/UX (Weeks 9–10):** Design and build responsive frontend interfaces for the Manager/Supervisor dashboard (`FR-11`), including interactive completion charts and cohort filter tables.
3. **Logistics Worker Usability Trials (Weeks 10–11):** Coordinate observational user testing sessions with target logistics workers, measuring task completion efficiency and recording qualitative UI feedback.
4. **Final Portfolio Visual Assets & Exhibition Deck (Week 12):** Produce the final enterprise product demonstration slide deck, demonstration video showreel, and final visual documentation manual.